Fiscal Year:	FY 2021	**Task Last Updated:**	FY 07/21/2022
PI Name:	Demmig-Adams, Barbara Ph.D.		
Project Title:	Co-Optimization of Duckweed Biomass, Nutritional Quality, and Input-Use Efficiency		
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Program/Discipline:			
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PI Email:	Barbara.Demmig-Adams@colorado.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	303-492-5541
Organization Name:	University of Colorado at Boulder		
PI Address 1:	Department of Ecology and Evolutionary Biology		
PI Address 2:	1900 Pleasant Street		
PI Web Page:			
City:	Boulder	**State:**	CO
Zip Code:	80309	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2018 TRA BRASH1801: Translational Research Institute for Space Health (TRISH) Biomedical Research Advances for Space Health
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Contact Monitor:		**Contact Phone:**	
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Adams, William Ph.D. (University of Colorado at Boulder)		
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Task Description:

This research effort proposes the use of small floating duckweeds as “multipurpose edible plants for spaceflight applications.” Duckweeds (family Lemnaceae) have many attractive characteristics for use as a space food crop. They are 100% edible, and can be consumed as a fresh, raw vegetable. They are among the fastest growing plants in the world, doubling their biomass in 1-3 days under ideal conditions.
Duckweeds are very small plants that are able to grow on thin films of still water. They can thus be grown on shallow, stacked trays, allowing for a high biomass yield per volume. Since they reproduce primarily through vegetative budding, pollination is not required. They can grow under a range of CO2 concentrations, continuously taking up CO2 through their permanently open leaf pores, and thus have a high capacity for cabin CO2 sequestration. Duckweeds are highly nutritious and hailed by some as the next super-food. Their dry mass is up to 45% high-quality protein that, like soy, provides all essential amino acids. They have a healthy ratio (<1) of essential omega-6 to omega-3 oils that is similar to flax seed. Duckweeds are also a good source of additional essential micronutrients, like beta carotene (provitamin A), vitamins C and E, and the antioxidant xanthophylls zeaxanthin and lutein, the combination of which protects the eye (and other organ systems) against radiation damage, which is particularly important for a crew exposed to space radiation. Chief among these protective compounds is zeaxanthin that is produced by plants only under specific light conditions and is thus often in limiting supply in the human diet.

In order to realize the plant’s full potential as a highly productive and nutritious crop, optimal growing conditions for production of high yields of nutritious food with the fewest spacecraft resources need to be defined in an environment relevant for space missions. Optimal light intensities – to maximize growth, light-use efficiency, and nutritional quality at elevated CO2 concentrations (up to 1%) – are not defined in literature. Also, high biomass yield often comes at the cost of poor micronutrient quality, and vice versa. A photosynthetic flux density (PFD) just enough to support maximal growth rate cannot be expected to induce a high vitamin/ antioxidant content for two principal reasons. Only exposure to light levels greater than what is needed for optimal growth will prompt leaves to accumulate antioxidants, which serve in defense against damage by excess light. Moreover, PFDs beyond those needed for optimal growth typically inhibit growth.

In this 2-year research effort, we propose a novel co-optimization of duckweed yield and micronutrient content by combining a growth rate-saturating continuous PFD with a small number of additional short, daily higher-light exposures, which generate a signal in the plant that stimulates antioxidant accumulation. We have already demonstrated proof-of-concept for this approach with a fast-growing weed, with successful co-optimization of biomass yield, micronutrient content, and light-use efficiency (Cohu CM et al., 2014).

The proposed study will utilize ground-based growth experiments to design a growth protocol that co-optimizes 1) edible biomass yield (and concomitant CO2 uptake), 2) protein content, 3) micronutrient content, and 4) energy efficiency (biomass/antioxidants produced per energy input) for two duckweed species at space-relevant CO2 concentrations up to 1%.

This project will be a collaborative effort between Dr. Barbara Demmig-Adams (Principal Investigator) and Dr. William Adams at the University of Colorado at Boulder; and Christine and Adam Escobar at Space Lab Technologies, LLC. Their combined facilities and expertise will provide a strong foundation for conducting the proposed research.

Reference:

Cohu CM, Lombardi E, Adams WW 3rd, Demmig-Adams B. Increased nutritional quality of plants for long-duration spaceflight missions through choice of plant variety and manipulation of growth conditions. Acta Astronautica 94(2), 799-806. 2014.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

This project combines innovations to increase plant nutrient content (protein and antioxidant micronutrients) for the human consumer while maintaining high volumetric yield and CO2 sequestration in duckweed biomass during a space mission.
1. Further identification of features that make duckweed plants an ideal space food crop with 100% edible biomass and an exceptionally high content of high-quality protein even under low growth light intensity, a high volumetric yield, and a superior capacity to accumulate the radiation-damage-fighting and systemic-inflammation-fighting micronutrient zeaxanthin.

2. Our mixed lighting protocol design – with low growth light and a pre-harvest finishing procedure with short-term exposure to high light – allows maximal growth (under low light by virtue of duckweed’s minimal self-shading and genetic propensity for unabated growth) combined with high-light induced production of zeaxanthin as an essential antioxidant/anti-inflammatory.

3. Novel non-destructive sensing options allow prediction of light-use efficiency of biomass production and zeaxanthin content.

4. Directions for the development of novel mitigation protocols (based on lowered growth light supply and/or plant-microbe interaction) that prevent adverse effects of long-term exposure to elevated CO2, including less protein, less zeaxanthin, lower growth rate as well as accelerated senescence.

These outcomes will not only fill a technology need for “multipurpose edible plants for spaceflight applications” (https://) but also establish a methodology for designing energy-efficient lighting protocols for production of nutrient-dense crops in controlled farming facilities. Duckweed naturally possesses many desirable traits for a space crop: rapid, continuous growth, 100% harvest index, high nutritional quality (up to 45% high-quality protein with all essential amino acids needed by humans, good potential source of vitamins A, C, and E and essential antioxidant/anti-inflammatories like zeaxanthin and lutein and omega-3 fats), tolerance to environmental extremes (including high CO2), and low infrastructure requirements.

Task Progress:	Our research goal was to design a specific growth lighting protocol suitable to co-optimize: (1) plant growth & CO2 uptake, (2) protein content, (3) micronutrient content, and (4) energy-use efficiency (biomass & antioxidants produced per energy input) for plants at space-relevant CO2 concentrations up to 1%. We developed small, fast-growing, floating duckweed plants (primarily of the genus Lemna) as a space crop that is 100% edible, grows well under high CO2 and microgravity, produces 20x more protein per area (for a single layer) than soybean and can, furthermore, be grown in trays stacked in multiple layers due to its diminutive size. In addition to protein content, we specifically optimized content of the essential human micronutrient zeaxanthin (a carotenoid) that fights radiation damage and systemic inflammation, and supports brain function. We determined that Lemna (and presumably other duckweeds) is unique among plants in: (i) producing exceptionally high concentrations of zeaxanthin in high light, (ii) growing equally fast in either low or high light, and (iii) quickly accumulating zeaxanthin upon sudden transfer of low-light-grown plants to high light for just 6 hours and without the high-light damage seen in other plants subjected to such a transfer. We traced these exceptional properties of Lemna to an unusual constitutive carotenoid composition, rapid acclimation to new growth conditions, unabated fast growth and high protein content under all growth light environments, and the absence of self-shading within the “canopy” of its single plant layer. These properties are particularly conducive for co-optimization via mixed lighting protocol consisting of continuous low growth light for a high light-use efficiency and a high content of high-quality protein (45% protein per dry biomass under 50 µmol photons m-2 s-1) combined with a short pre-harvest finishing exposure to high light to produce high levels of radiation-damage-fighting zeaxanthin. We prepared and published several reviews on the effect of zeaxanthin as a unique radiation-damage-fighting molecule for plants and humans. Moreover, we identified approaches for remote sensing of plant biomass production and zeaxanthin level. We also identified challenges to the growth of duckweed under elevated CO2, as also seen in other species, which included decreased protein and zeaxanthin contents as well as accelerated senescence and lower growth rates under some conditions. We conducted a comprehensive literature review of plant response to high CO2, and concluded that high CO2 – especially in combination with other conditions that affect the balance between sugar production and consumption – can cause metabolic disruption. We hypothesized that this metabolic disruption may be prevented by plant-microbe interaction, where microbes serve as an additional sugar sink and provide plant growth regulators. We were able to obtain preliminary support for this hypothesis by inoculating sanitized Lemna with microbial partners, which prevented declines in protein content and growth rate under elevated CO2. This insight provides a further reason for growing plants under low to moderate (rather than high) light intensity under conditions of elevated CO2, and makes our newly developed mixed lighting protocol particularly useful (especially if maintaining a sterile environment during manned spaceflight continues to be a priority).

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